

RTutor.ai Quick-Start Guide

Scatterplots, Trend Lines, and Multivariable Scatterplots
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0) Setup

Use the same Exam Grades dataset from the previous guides. If you are starting fresh:

1. Upload the dataset in RTutor.ai under **(1. Data)** → **Browse**.
2. Confirm four columns: *Student ID*, *Year In School*, *Practice Exam Grade*, *Actual Exam Grade*.
3. Under **(2. Data Types)** → **Edit**, set Student ID = Character, Year In School = Category, both grade columns = Integer (or Numeric).

1) Scatterplot

Goal: Create a scatterplot with Practice Exam Grade on the x-axis and Actual Exam Grade on the y-axis.

Your prompt to RTutor.ai must:

1. Reference the Exam Grades dataset and specify both variables: Practice Exam Grade (x-axis) and Actual Exam Grade (y-axis).
2. Say “scatterplot” (not bar graph, not line graph).
3. Set both axes from 0 to 100 with tick marks every 10 points. Request font size 14 and a color of your choice for the points.

Check: Is Practice Exam Grade on the x-axis and Actual Exam Grade on the y-axis? Are individual points (not bars or lines) displayed? Do both axes run 0 to 100?

2) Trend Line

Goal: Add a linear trend line (best-fit line) to the scatterplot.

Your prompt must include:

1. Ask RTutor.ai to recreate the scatterplot from Section 1 and add a linear trend line (best-fit line). If RTutor.ai adds a curved line, specify “linear” or “straight line.”
2. Request that the trend line be a distinct color from the points so it is easy to see.

Check: Is there a single straight line overlaid on the scatterplot? Does the line’s direction make sense (positive slope if higher practice grades tend to go with higher actual grades)?

3) Multivariable Scatterplot

Goal: Recreate the scatterplot with points colored by Year In School, and add a separate trend line for each group.

Your prompt must include:

1. Specify Practice Exam Grade (x-axis), Actual Exam Grade (y-axis), and color or shape the points by Year In School.
2. Ask for a separate linear trend line for each Year In School group, each in the same color as its points.
3. Keep both axes 0 to 100, tick marks every 10, font size 14, and place the legend at the bottom.

Check: Can you distinguish Freshman points from Upperclassman points by color or shape? Are there two separate trend lines? Does the legend clearly identify each group?

4) Interpretation

1. Write 2–3 sentences about the overall scatterplot: Describe the direction (positive, negative, or none) and strength of the relationship between Practice Exam Grade and Actual Exam Grade. Note whether the trend line confirms this.
2. Write 2–3 sentences about the multivariable scatterplot: Compare the two groups' trend lines. Do Freshmen and Upperclassmen show a similar relationship between practice and actual grades, or does one group have a steeper or shifted line?

5) Troubleshooting

1. **Wrong variable / not found:** Use exact column names (Student ID, Year In School, Practice Exam Grade, Actual Exam Grade).
2. **Wrong graph type:** Say “scatterplot” explicitly, and add “not a bar graph” or “not a line graph” if needed.
3. **Axes swapped:** Explicitly state which variable is on the x-axis and which is on the y-axis in your prompt.
4. **Curved trend line instead of straight:** Specify “linear trend line” or “straight best-fit line.”
5. **Groups not distinguished:** Make sure Year In School is set to Category. Say “color by Year In School” or “shape by Year In School.”
6. **Prompt partially ignored:** Break long prompts into smaller steps: basic scatterplot → trend line → color by group → group trend lines.

Still stuck? Save your prompt and a screenshot, then ask a CA or instructor.